Affine maps carry Gaussian means and covariances

$$x \in \mathbb{R}^d, \quad y = Ax + b \in \mathbb{R}^m$$

$$A \in \mathbb{R}^{m \times d}, \quad b \in \mathbb{R}^m \quad \text{(deterministic)}$$

$$E[x] = \mu, \quad \mathrm{Cov}[x] = \Sigma$$

$$E\left[\sum_{i=1}^{d} x_i^2\right] < \infty$$

Moment proof

$$\mu_y = E[y] = AE[x] + b = A\mu + b$$

$$y - \mu_y = Ax + b - (A\mu + b) = A(x - \mu)$$

$$\begin{aligned}
\Sigma_y &= E\left[(y - \mu_y)(y - \mu_y)^T\right] \\
&= E[A(x - \mu)(x - \mu)^T A^T] \\
&= AE[(x - \mu)(x - \mu)^T]A^T \\
&= A\Sigma A^T
\end{aligned}$$

- Translation shifts the Gaussian cloud's mean without changing its covariance.
- A linearly maps its center and spread; affine closure keeps the transformed cloud Gaussian.